

Cognitive AI Project - New Llama 3 Based Architecture Plan

1. Project Goal: Build a Cognitive Decision Support System for career and life decision-making with stronger human-like personalized guidance.
2. Main Upgrade: Replace FLAN-T5 with Meta Llama 3 (8B Instruct) using Google Colab GPU and LoRA fine-tuning.
3. Python Strategy: Use Python 3.14 locally for Streamlit UI, ML, NLP, simulation support, and overall project execution. Use Google Colab only for Llama 3 training.
4. Core Modules: ML classification, NLP preprocessing, emotion detection, future simulation, weekly planner, Llama 3 personalized advice generation, and Streamlit conversational UI.
5. Recommended Hybrid Flow: ML predicts category -> NLP detects emotional state -> Llama 3 generates planner + simulation + mentor-style final advice.
6. Why Llama 3: Better emotional understanding, better counseling-style output, stronger personalized guidance, better viva/project presentation quality.
7. Planner Upgrade: Replace static rule-based weekly planner with dynamic Llama-generated personalized weekly action plans.
8. Simulation Upgrade: Replace fixed simulation logic with Llama-generated short-term and long-term outcome analysis based on user context.
9. Training Method: Use LoRA fine-tuning instead of full training for faster and cheaper training on Colab GPU.
10. Final Folder: Save trained model inside `llm/llama_model/` and load it in Streamlit app locally.